Course Outline for CSE-436

Part A

1. Course Code: CSE-436

2. Course Title: Information Security (Sessional)

3. Course Type: Core Course

4.Level/ Term: Level: 4 Term: I

5. Academic Session: 2020-21

6. Course Teacher: Dr. Md. Mokammel Haque, Professor, Dept. of CSE, CUET and
Ashim Dey, Assistant Professor, CSE, CUET

7. Prerequisite(s): None

8. Credits: 0.75 (3 hours of lab work every fortnight)

9. Contact Hours: 3 hours of lab work per fortnight/alternate week

10. Total Marks: 75

11. Rational of the Course:

This Lab/Sessional course will cover hands-on experience on managing different types of cyberattacks, vulnerability analysis and prevention strategies. This course is important to understand the fundamental of information security technology concept. This is a required course for all the students enrolling B. Sc. Engg. in CSE program from session 2020-2021. The catalogue description of the course is

Course Content:

Sessional based on the following topics:
Concept and applications of cryptography; Techniques of designing efficient Cryptosystems: Symmetric key encryption- Stream ciphers, Block ciphers; Message authentication codes; Hash function; Digital signature; Asymmetric key encryption- Diffie-Hellman protocol, Trapdoor functions, RSA, Merkel puzzles, El-Gammal cryptosystems; Linear and differential cryptanalysis; Mathematics of cryptography; Lightweight cryptographic protocols; Steganography.

Fundamental properties and terminology of information security; System security management, analysis and control; Physical and logical security; Types of attacks; Database security; Network security threats; WSN security; Fundamental of Cybersecurity; IoT network and IoT device security; Blockchain; Secure Cloud computing; Computer abuse; Legal and Ethical issues.

12. Course Objectives:

(a) To understand various types of security threats and prevention strategies

(b) To learn how to cryptanalyze or attack a cryptosystem

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

No.	Course Learning Outcomes (CLOs)	Bloom's Level (Optional)
CLO1	Analyze different types of threats and malicious attacks	C1, P1, P2, P5
CLO2	Cryptanalyze a cryptosystem	C2, P1, P2, P5

b) Mapping of CLOs with PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PL11	PL12
CLO1	X	X			X							
CLO2	X	X			X							

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Course Plan

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Basic of Security and Hacking -Why system get hacked & Practical discussion on how to hack and protect	• Lecture on theoretical background • Hands on demonstration • Group Discussion	Lab Performance Report	• CLO-1
Week -03	Brute-force attack and Man-in-the	• Lecture on theoretical background	Lab Performance	• CLO-2

	Middle attack on Classical crypto	• Hands on demonstration • Group Discussion	Report	
Week - 05	Vulnerability Analysis, Sniffing in real-life action and hacking a Web Server	• Lecture on theoretical background • Hands on demonstration • Group Discussion	Lab Performance Report	• CLO-1
Week - 07	SQL Injection in Real life and Session Hijacking	• Lecture on theoretical background • Hands on demonstration • Group Discussion	Lab Performance Report	• CLO-1
Week - 09	Data destruction, Data backup, recovery technique and infected files manipulation using Sandbox	• Lecture on theoretical background • Hands on demonstration • Group Discussion	Lab Performance Report	• CLO-1
Week - 11	Cybersecurity Data Analytics and Intelligent decision making. Cyber Attacks, Anomaly detection & Business risk prediction	• Lecture on theoretical background • Hands on demonstration • Group Discussion	Lab Performance Report	• CLO-1
Week - 12	Lab Test			
Week - 13	Viva-voce/Quiz			

Part C

15. Assessment and Evaluation

1) Assessment Strategy:

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%

b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

Charles P. Pfleeger et al., Security in Computing, Prentice-Hall.

2) Supplementary Readings

Paar, Christof, et al, Understanding Cryptography, Springer

3) Others

Laboratory Manuel (if any) for CSE-436